

Notes: Cox & Shi (RES, 2023)

Simple Adaptive Size-Exact Testing for Full-Vector and Subvector Inference in Moment Inequality Models

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Model setup and measurement objects	2
3.1	Full-vector moment inequality model	2
3.2	Likelihood-ratio distance	2
3.3	Active rank and CC critical value	3
3.4	Subvector setup	3
4	Tests, algorithms, and theoretical properties	3
5	Identification and theoretical design logic	3
6	Tables, figures, and algorithms	3
6.1	Figure 1: Geometric Representation of the CC Test and the RCC Test + Algorithm 1: Pseudo-code for Implementing the CC Test	3
6.2	Algorithm 2: Pseudo-code for the sCC Test Using Vertex Enumeration of H	4
6.3	Table 1: Finite Sample Maximum Null Rejection Probabilities and Size-corrected Average Power, Known Ω	4
6.4	Table 2: Finite Sample Maximum Null Rejection Probabilities and Power, Estimated Ω	5
6.5	Figure 2: Power Curves for Five Tests	5
6.6	Table 3: Large k and n Cases	5
6.7	Table 4: Confidence Interval Value, Length, and Computation Time	6
6.8	Figure 3: Rejection Rates of the sCC, sRCC, and ARP Hybrid Tests	6
7	Result map	6
8	Short summary	7
9	Conclusion	7
9.1	Core conclusion	7
9.2	Intuition	7
9.3	Incremental contribution	8
9.4	Limitations	8
9.5	Takeaway	8

1 Big picture

- **Question.** How can researchers conduct valid and computationally cheap inference in moment inequality models when confidence sets are constructed by testing many parameter values?
- **Problem.** Standard moment-inequality inference often requires simulated critical values, moment-selection tuning parameters, or searches over high-dimensional nuisance parameters.
- **Main contribution.** The paper proposes a conditional chi-squared (CC) test whose critical value is a chi-squared quantile with degrees of freedom equal to the rank of active inequalities at the restricted estimator.
- **Core intuition.** Slack inequalities should not count toward the effective dimension of the binding problem.
- **Practical appeal.** The test is simulation free, tuning-parameter free, easy to code, and fast enough for test inversion over dense parameter grids.

2 Incremental contribution of the paper

- **Relative to bootstrap/simulation methods:** it replaces simulated critical values with analytical chi-squared critical values.
- **Relative to moment-selection methods:** it adapts to slackness without tuning thresholds.
- **Relative to full-vector-only inference:** it adds a subvector procedure when nuisance parameters enter linearly.
- **Relative to applied bottlenecks:** it reduces both per-test cost and the dimension of grids needed for confidence-set construction.
- **Relative to earlier theory:** it combines exact finite-sample size in normal models with uniform asymptotic validity and slackness adaptation.

3 Model setup and measurement objects

3.1 Full-vector moment inequality model

$$AE_F m_n(\theta) \leq b, \quad m_n(\theta) = n^{-1} \sum_i m(W_i, \theta).$$

$$\Theta_0(F) = \{\theta \in \Theta : AE_F m_n(\theta) \leq b\}.$$

Detailed interpretation: The framework includes equality moments by writing them as pairs of inequalities. The paper therefore covers a broad class of partial-identification models.

3.2 Likelihood-ratio distance

$$T_n(\theta) = \min_{\mu: A\mu \leq b} n(m_n(\theta) - \mu)' \hat{\Sigma}_n(\theta)^{-1} (m_n(\theta) - \mu).$$

Detailed interpretation: This statistic measures how far the sample moments are from satisfying the restrictions at candidate parameter value θ .

3.3 Active rank and CC critical value

Let $\hat{\mu}$ solve the restricted problem. Define the active set $\hat{J} = \{j : a'_j \hat{\mu} = b_j\}$ and active rank $\hat{r} = \text{rank}(A_{\hat{J}})$. The CC test rejects when

$$T_n(\theta) > \chi_{\hat{r}, 1-\alpha}^2.$$

Detailed interpretation: Rank rather than number of active inequalities is used so that redundant inequalities do not inflate the critical value.

3.4 Subvector setup

For conditional moment inequalities with linearly entering nuisance parameters,

$$H_0 : \exists \delta \text{ such that } B_Z E[m_n(\theta_0)|Z] - C_Z \delta \leq d_Z.$$

The test eliminates δ to obtain an equivalent inequality in θ alone.

Detailed interpretation: This reduces the dimension of test inversion from (θ, δ) to θ .

4 Tests, algorithms, and theoretical properties

- The CC test is simulation free and tuning-parameter free.
- The refined CC test improves exact finite-sample size properties in normal models.
- The tests are uniformly asymptotically valid under asymptotic normality.
- The tests adapt to slackness because slack inequalities usually do not become active at the restricted estimator.
- The subvector version uses convex geometry and linear programming ideas to eliminate nuisance parameters.

5 Identification and theoretical design logic

This is a methods paper. The relevant identification object is the set of parameters satisfying moment inequalities. The key difficulty is that the distribution of the likelihood-ratio statistic depends on which inequalities are binding. The CC test solves this by using the active rank of the binding constraints.

6 Tables, figures, and algorithms

6.1 Figure 1: Geometric Representation of the CC Test and the RCC Test + Algorithm 1: Pseudo-code for Implementing the CC Test

Read: Figure 1 shows the CC and RCC rejection regions; Algorithm 1 shows the implementation steps.

Detailed interpretation: The local face of the feasible moment set determines the active rank and therefore the critical value.

Economic message: The test converts moment inequality inference into a geometry-aware chi-squared comparison.

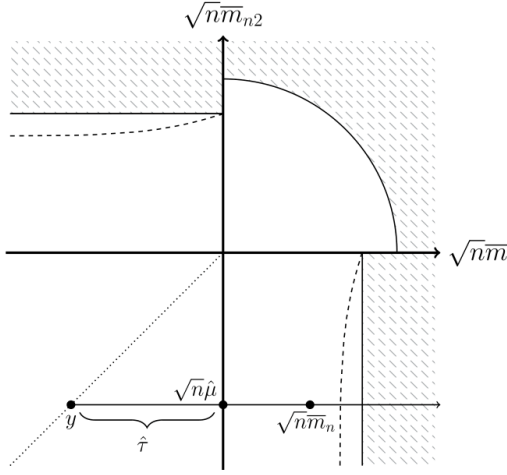


FIGURE 1

ometric representation of the CC test (shaded) and the RCC test (dashed) in III

Algorithm 1: Pseudo-code for implementing the CC test.

```

1: % Compute the CC Test
2:  $T_n(\theta), \hat{\mu} \leftarrow \min_{\mu: A\mu \leq b} n(\bar{m}_n(\theta) - \mu)' \hat{\Sigma}_n(\theta)^{-1} (\bar{m}_n(\theta) - \mu)$ 
3:  $\hat{J} := \{j = 1, \dots, d_A : a_j' \hat{\mu} = b_j\}$ 
4:  $A_{\hat{J}} \leftarrow \hat{J}, A$ 
5:  $\hat{r} := \text{rk}(A_{\hat{J}})$ 
6:  $\phi_n^{\text{CC}}(\theta, \alpha) := 1\{T_n(\theta) > \max\{\chi_{\hat{r}, 1-\alpha}^2, \text{tol}\}\}.$ 

```

(a) Figure 1: CC and RCC geometry

(b) Algorithm 1: CC test

6.2 Algorithm 2: Pseudo-code for the sCC Test Using Vertex Enumeration of $\mathcal{H}$

Read: Algorithm 2 extends CC testing to subvector inference by eliminating nuisance parameters.

Detailed interpretation: The procedure avoids gridding over nuisance parameters when they enter linearly.

Economic message: This is the main applied gain for empirical models with many controls or nuisance terms.

Algorithm 2: Pseudo-code for the sCC test using vertex enumeration of $\mathcal{H}$.

```

1: % Compute the sCC Test
2:  $T_n(\theta), \hat{\mu} \leftarrow \min_{\delta, \mu: B_Z \mu - C_Z \delta \leq d_Z} n(\bar{m}_n(\theta) - \mu)' \hat{\Sigma}_n(\theta)^{-1} (\bar{m}_n(\theta) - \mu)$ 
3:  $H(C_Z) \leftarrow \mathcal{H}$  using a vertex enumeration algorithm, e.g. con2vert.m in Matlab (ref. Kleider (2020))
4:  $A_Z, b_Z \leftarrow H(C_Z) B_Z, H(C_Z) d_Z$ 
5:  $\hat{J} := \{j = 1, \dots, d_A : e_j' A_Z \hat{\mu} = e_j' b_Z\}$ 
6:  $\hat{r} := \text{rk}(I_{\hat{J}} A_Z)$ 
7:  $\phi_n^{\text{sCC}}(\theta, \alpha) := 1\{T_n(\theta) > \max\{\chi_{\hat{r}, 1-\alpha}^2, \text{tol}\}\}.$ 

```

Algorithm 2: sCC test

6.3 Table 1: Finite Sample Maximum Null Rejection Probabilities and Size-corrected Average Power, Known Ω

Read: CC/RCC have MNRP near nominal size and are dramatically faster than bootstrap-based alternatives.

Detailed interpretation: The table demonstrates the full-vector test in the clean known-variance case.

Economic message: The computation-time columns explain why the method is useful for test inversion.

TABLE 1
Finite sample maximum null rejection probabilities and size-corrected average power of nominal 5% tests (normal distribution, known Ω , $n=100$)

Test	$k=10$				$k=4$				$k=2$			
	MNRP	WAP	ScWAP	Time	MNRP	WAP	ScWAP	Time	MNRP	WAP	ScWAP	Time
$\Omega = \Omega_{\text{Neg}}$												
RCC	0.051	0.61	0.61	0.003	0.052	0.62	0.62	0.003	0.051	0.62	0.62	0.003
CC	0.051	0.61	0.60	0.003	0.049	0.60	0.61	0.003	0.046	0.58	0.60	0.003
AB	0.046	0.53	0.55	1.11	0.051	0.59	0.59	1.07	0.059	0.65	0.64	1.40
RSW1	0.054	0.58	0.56	0.551	0.056	0.60	0.59	0.538	0.052	0.64	0.63	0.701
RSW2	0.050	0.23	0.23	0.014	0.052	0.34	0.34	0.013	0.052	0.50	0.49	0.014
$\Omega = \Omega_{\text{Zero}}$												
RCC	0.052	0.63	0.63	0.003	0.052	0.65	0.65	0.003	0.051	0.68	0.68	0.003
CC	0.050	0.62	0.62	0.003	0.045	0.62	0.64	0.003	0.038	0.61	0.66	0.004
AB	0.043	0.65	0.66	1.08	0.050	0.67	0.67	1.07	0.056	0.69	0.67	1.39
RSW1	0.053	0.61	0.60	0.545	0.056	0.63	0.62	0.539	0.052	0.65	0.65	0.699
RSW2	0.053	0.54	0.52	0.014	0.052	0.62	0.62	0.014	0.049	0.66	0.66	0.014
$\Omega = \Omega_{\text{Pos}}$												
RCC	0.051	0.76	0.75	0.003	0.053	0.75	0.74	0.003	0.051	0.72	0.71	0.003
CC	0.038	0.72	0.76	0.003	0.033	0.68	0.74	0.003	0.032	0.62	0.69	0.003
AB	0.042	0.78	0.80	1.05	0.051	0.75	0.75	1.03	0.059	0.72	0.70	1.34
RSW1	0.053	0.77	0.77	0.547	0.056	0.73	0.71	0.534	0.052	0.67	0.66	0.700
RSW2	0.052	0.77	0.77	0.014	0.052	0.74	0.74	0.013	0.049	0.68	0.69	0.014

Notes: CC, RCC, AB, RSW1, and RSW2 denote the CC test, the refined CC test, the AQLR test with bootstrap critical values in AB, the two-step test in RSW1 based on the GPD statistic, and that based on the Max statistic, respectively. MNRP

TABLE 2
Finite sample maximum null rejection probabilities and size-corrected average power of nominal 5% tests (normal distribution, estimated Ω , $n=100$)

Test	$k=10$				$k=4$				$k=2$			
	MNRP	WAP	ScWAP	Time	MNRP	WAP	ScWAP	Time	MNRP	WAP	ScWAP	Time
$\Omega = \Omega_{\text{Neg}}$												
RCC	0.074	0.63	0.54	0.003	0.058	0.63	0.058	0.61	0.004	0.053	0.62	0.003
CC	0.074	0.63	0.54	0.003	0.058	0.61	0.59	0.004	0.048	0.59	0.60	0.003
AB	0.046	0.51	0.53	1.23	0.049	0.58	0.58	1.56	0.056	0.64	0.63	1.13
RSW1	0.054	0.55	0.53	0.569	0.053	0.58	0.58	0.736	0.050	0.62	0.62	0.533
RSW2	0.053	0.24	0.23	0.026	0.051	0.34	0.33	0.026	0.051	0.49	0.48	0.023
$\Omega = \Omega_{\text{Zero}}$												
RCC	0.069	0.65	0.59	0.003	0.053	0.66	0.65	0.004	0.051	0.68	0.68	0.003
CC	0.069	0.64	0.57	0.003	0.049	0.63	0.63	0.004	0.039	0.61	0.66	0.003
AB	0.043	0.62	0.64	1.23	0.048	0.66	0.67	1.55	0.053	0.68	0.67	1.14
RSW1	0.052	0.58	0.58	0.570	0.053	0.62	0.61	0.732	0.050	0.64	0.64	0.541
RSW2	0.062	0.54	0.50	0.026	0.053	0.61	0.60	0.026	0.050	0.64	0.64	0.024
$\Omega = \Omega_{\text{Pos}}$												
RCC	0.056	0.77	0.75	0.003	0.054	0.75	0.74	0.004	0.051	0.71	0.71	0.003
CC	0.043	0.73	0.74	0.003	0.034	0.68	0.73	0.004	0.035	0.63	0.69	0.003
AB	0.044	0.78	0.79	1.19	0.049	0.74	0.75	1.50	0.055	0.71	0.70	1.07
RSW1	0.053	0.76	0.75	0.566	0.053	0.71	0.71	0.731	0.052	0.66	0.66	0.528
RSW2	0.056	0.76	0.74	0.026	0.052	0.73	0.72	0.027	0.050	0.67	0.67	0.023

Notes: CC, RCC, AB, RSW1, and RSW2 denote the CC test, the refined CC test, the AQLR test with bootstrap critical values in AB, the two-step test in RSW1 based on the GPD statistic, and that based on the Max statistic, respectively. MNRP

Table 1: Known Ω simulation results

Table 2: Estimated Ω simulation results

6.4 Table 2: Finite Sample Maximum Null Rejection Probabilities and Power, Estimated Ω

Read: The table repeats Table 1 with estimated covariance matrices.

Detailed interpretation: The method remains competitive in a more realistic applied setting.

Economic message: Variance estimation changes finite-sample behaviour but not the core computational advantage.

6.5 Figure 2: Power Curves for Five Tests

Read: The panels show that different tests allocate power differently over violation directions.

Detailed interpretation: No test is uniformly most powerful, but CC/RCC are competitive across directions.

Economic message: The simple test does not buy speed by becoming powerless.

6.6 Table 3: Large k and n Cases

Read: CC/RCC remain computationally cheap when the number of inequalities grows.

Detailed interpretation: The table is encouraging for scalability but highlights challenges under non-normality.

Economic message: The method expands the feasible set of moment-inequality applications.

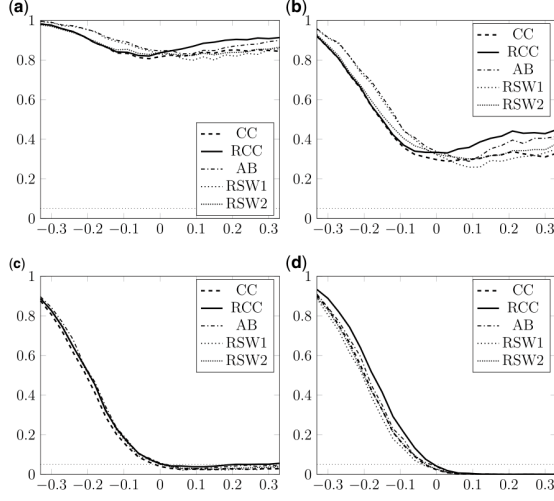


FIGURE 2
Power curves for five tests ($k=2$, normal distribution $\Omega=\Omega_{Z_{\text{max}}}$, estimated Ω , $n=100$, $\alpha=5\%$)

Figure 2: Power curves

TABLE 3
Finite sample maximum null rejection probabilities of nominal 5% tests, the large k and n cases (estimated Ω , $\Omega=\Omega_{Z_{\text{max}}}$)

Test	$k=10, n=100$		$k=50, n=700$		$k=100, n=1600$		$k=150, n=2550$	
	MNRP	Time	MNRP	Time	MNRP	Time	MNRP	Time
Normal								
RCC	0.069	0.003	0.074	0.004	0.076	0.011	0.081	0.024
CC	0.069	0.003	0.074	0.005	0.076	0.011	0.081	0.024
RSW1	0.052	0.582	0.062	1.07	0.047	3.25	0.051	7.96
RSW2	0.062	0.027	0.056	0.211	0.048	1.15	0.045	4.04
$t(3)$								
RCC	0.063	0.003	0.069	0.004	0.071	0.010	0.079	0.024
CC	0.063	0.003	0.069	0.005	0.071	0.011	0.079	0.024
RSW1	0.057	0.568	0.054	1.07	0.050	3.23	0.054	7.97
RSW2	0.093	0.026	0.069	0.210	0.061	1.24	0.063	4.05
Mixed normal								
RCC	0.096	0.003	0.090	0.004	0.089	0.010	0.089	0.024
CC	0.096	0.003	0.090	0.005	0.089	0.010	0.089	0.024
RSW1	0.045	0.567	0.051	1.06	0.059	3.20	0.054	7.92
RSW2	0.086	0.026	0.069	0.210	0.065	1.24	0.058	4.04

Notes: CC, conditional chi-squared test; RCC, refined CC test; RSW1 and RSW2, two-step test in RSW based on the QLR statistic and the Max statistic; MNRP, maximum null rejection probability; Time, average computation time in seconds for the test in each Monte Carlo repetition. The RSW tests use 499 critical value simulations. The results for the CC, RCC

Table 3: Large k, n cases

6.7 Table 4: Confidence Interval Value, Length, and Computation Time

Read: sCC/sRCC intervals are shorter and much faster than the ARP hybrid intervals in the reported design.

Detailed interpretation: The table shows the payoff from nuisance-parameter elimination and analytical critical values.

Economic message: Subvector CC is the paper's strongest practical tool for applied confidence intervals.

6.8 Figure 3: Rejection Rates of the sCC, sRCC, and ARP Hybrid Tests

Read: Rejection rates are at or below 5% inside the identified set and rise outside the set.

Detailed interpretation: The figure confirms size control and shows how power increases away from the identified set.

Economic message: The subvector confidence intervals are valid and informative.

7 Result map

- The CC statistic is a likelihood-ratio distance to the feasible set.
- The critical value uses the rank of active restrictions.
- The refined version obtains exact finite-sample size in normal models.
- The subvector procedure eliminates linearly entering nuisance parameters.
- Simulations show large computational gains and competitive size and power.

TABLE 4
Average value, length, and computation time (in seconds) of confidence intervals

n=500			n=1000			
CI	Excess length	Time	CI	Excess length	Time	
$d_c = 2, 8$ moment inequalities						
sRCC	$[-1.774, -0.339]$	0.989	2.5	$[-1.609, -0.440]$	0.723	2.4
sCC	$[-1.780, -0.332]$	1.00	2.5	$[-1.615, -0.433]$	0.736	2.4
ARP hybrid	$[-1.998, -0.264]$	1.29	111	$[-1.736, -0.395]$	0.895	109
$d_c = 3, 16$ moment inequalities						
sRCC	$[-1.852, -0.293]$	1.11	7.1	$[-1.659, -0.404]$	0.809	4.7
sCC	$[-1.852, -0.293]$	1.11	7.1	$[-1.659, -0.404]$	0.809	4.7
ARP hybrid	$[-2.219, -0.123]$	1.65	199	$[-1.883, -0.287]$	1.15	120
$d_c = 4, 32$ moment inequalities						
sRCC	$[-1.921, -0.254]$	1.22	6.5	$[-1.718, -0.366]$	0.906	10
sCC	$[-1.921, -0.254]$	1.22	6.5	$[-1.718, -0.366]$	0.906	10
ARP hybrid	$[-2.596, -0.011]$	2.14	97	$[-2.104, -0.180]$	1.48	145

Notes: The identified set for θ_0 is $[-1.203, -0.757]$. The computation times across different (n, d_c) cases are not comparable because they may have been performed by different computers on the computer cluster. The computation of different tests within each (n, d_c) case is always completed on the same computer. Thus the computation times across

Table 4: Subvector confidence intervals

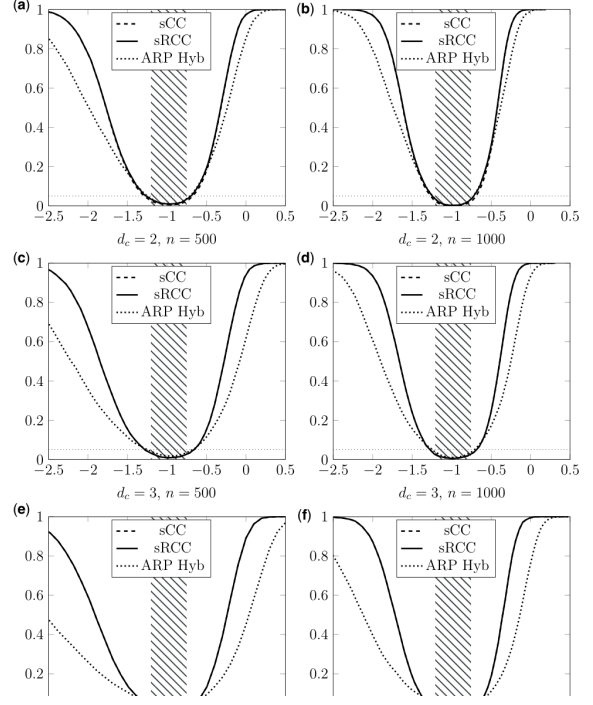


Figure 3: Subvector rejection rates

8 Short summary

- Cox and Shi propose a simple adaptive test for moment inequality models.
- The test uses chi-squared critical values with active-rank degrees of freedom.
- It is simulation free, tuning-parameter free, and fast to compute.
- It adapts to slack inequalities without explicit moment selection.
- It extends to subvector inference with linearly entering nuisance parameters.
- Simulations show strong computational advantages and competitive finite-sample performance.

9 Conclusion

9.1 Core conclusion

Cox and Shi show that moment inequality inference can be valid, adaptive, and computationally simple. The active-rank chi-squared critical value is the central insight.

9.2 Intuition

Only the binding face of the feasible set determines the local distribution of the likelihood-ratio statistic. Slack constraints should not count toward the degrees of freedom.

9.3 Incremental contribution

The paper eliminates simulation and tuning parameters, adds subvector inference through nuisance elimination, and provides algorithms that are practical for confidence-set construction.

9.4 Limitations

The main theory relies on normal or asymptotically normal approximations. High-dimensional many-inequality theory and difficult covariance estimation remain important future work.

9.5 Takeaway

The paper makes moment inequality inference easier to implement by combining convex geometry with simple chi-squared critical values.